

EXPLORING THE EFFECTS OF CURCUMIN AND NIMBOLIDE ON B16-F10 MELANOMA CELL VIABILITY

Introduction

Melanoma Skin Cancer

Melanoma is a **highly metastatic** skin cancer that originates in **melanocytes** (Pennmedicine., 2025). It is the most aggressive form of skin cancer. Melanoma's primary cause is **UV radiation** from sunlight and indoor tanning beds (National Cancer Society., 2026).

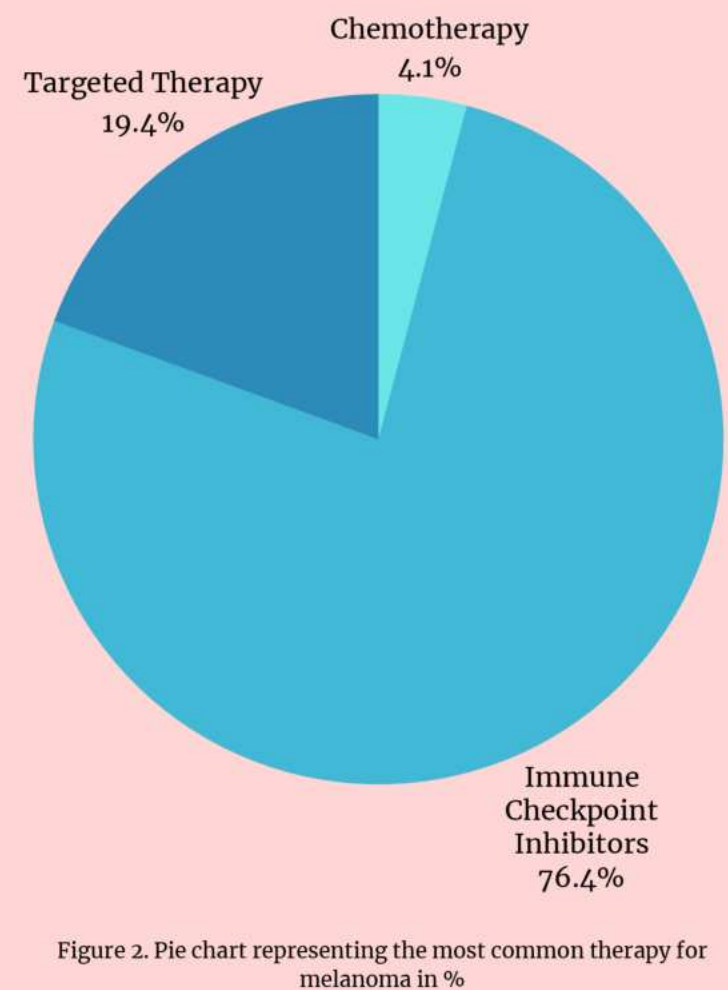


Figure 2. Pie chart representing the most common therapy for melanoma in %



Once melanoma metastasizes, 1 out of 3 people are expected to survive within 5 years. (Skin Cancer Foundation., 2021)

Figure 1. 5 year survival rate once melanoma metastizes, indicating poor prognosis

Immune Checkpoint Inhibitors boost the body's immune system to kill cancer cells. Side effects of the treatment can cause the immune system to target healthy organs (Chhabra et al., 2021).

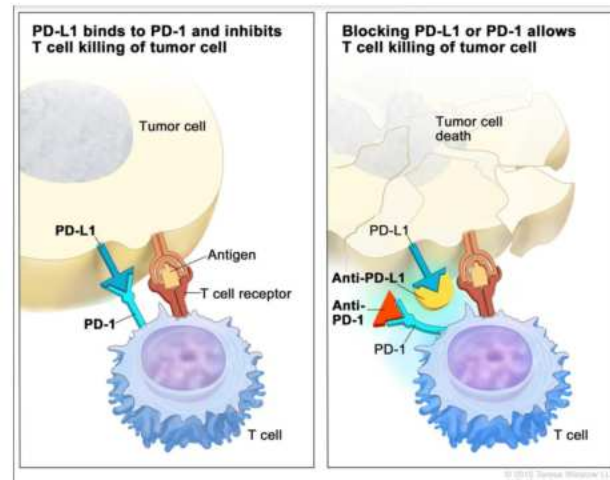


Figure 3. Visualization of Immune Checkpoint Inhibition. Demonstrates method helper T-Cells use to identify cancer cells.

Project Overview

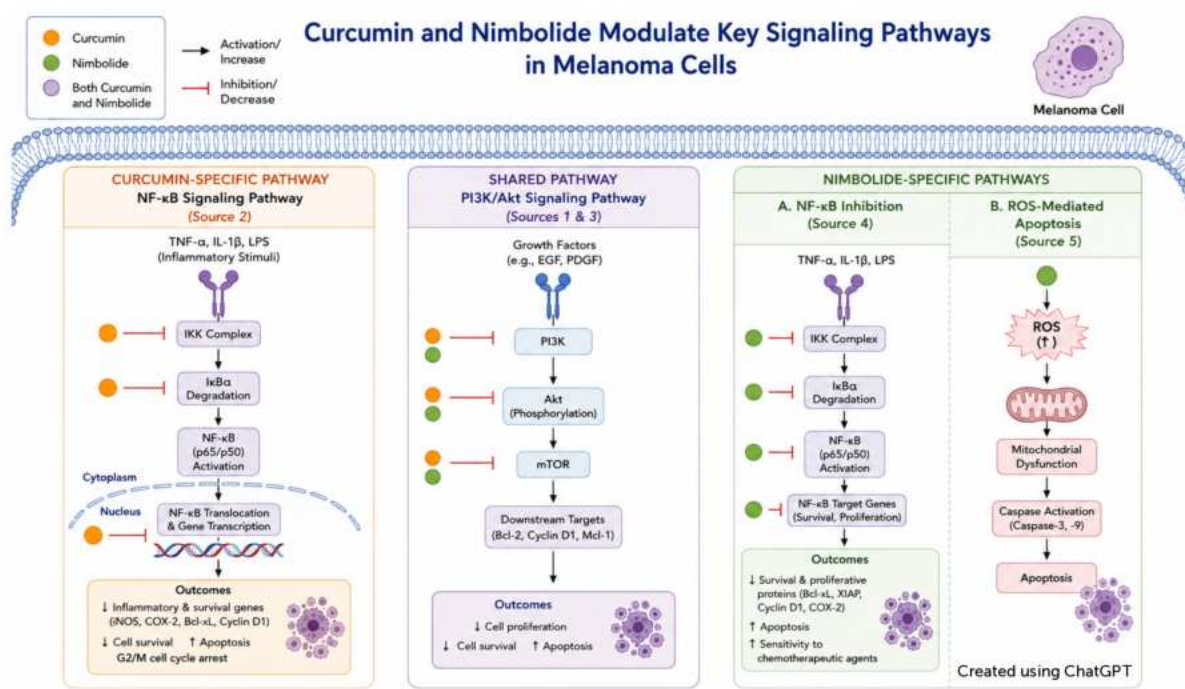


Figure 4. Visualization of Oncogenic Signaling Inhibition by Curcumin and Nimbolide

Research Question

What are the effects of curcumin and nimbolide on cell viability in the B16-F10 Melanoma Mouse Cell Line?

Hypothesis

If B16-F10 cells are treated with both Curcumin and Nimbolide, then cell viability will decrease synergistically.

Methodology

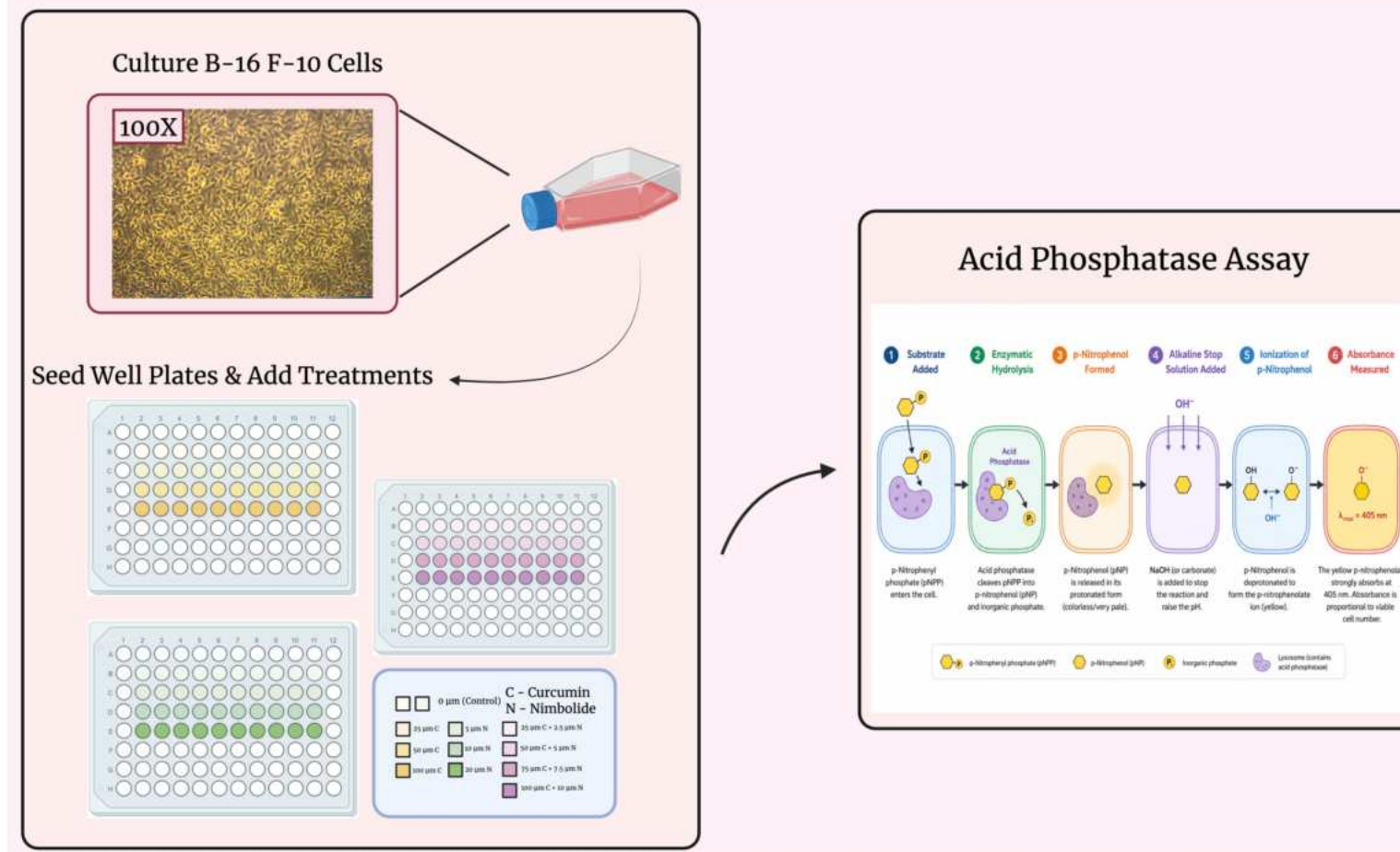


Figure 5. Visualization of Methods used in Study
Graphic created by researcher using Biorender (2025)

Results

Curcumin Treatment

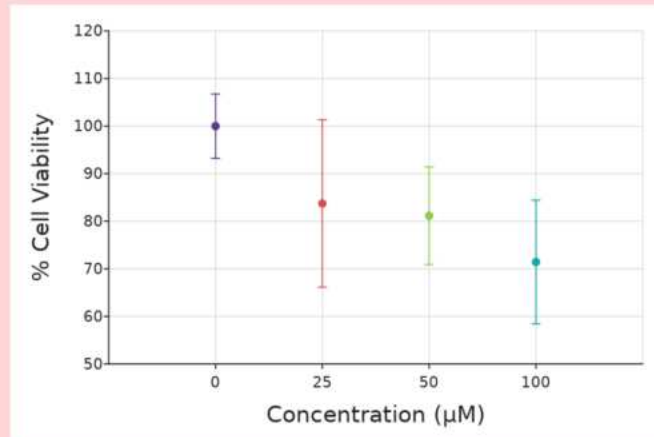


Figure 6. Graph representing cell viability in relation to curcumin dose concentration. Treatment groups were normalized to the control.

Nimbolide Treatment

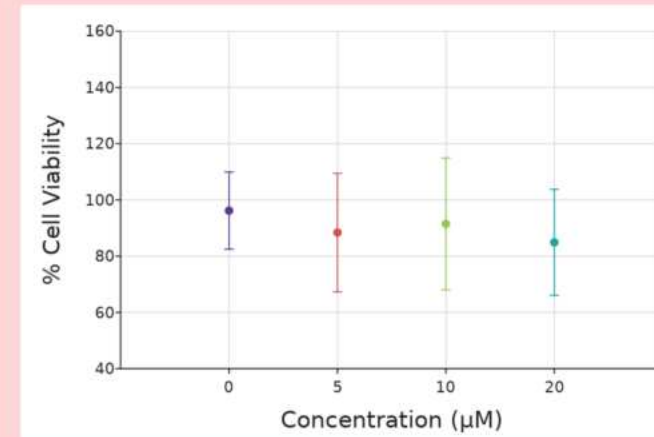


Figure 7. Graph representing cell viability in relation to nimbolide dose concentration. Treatment groups were normalized to the control.

Combined Treatment

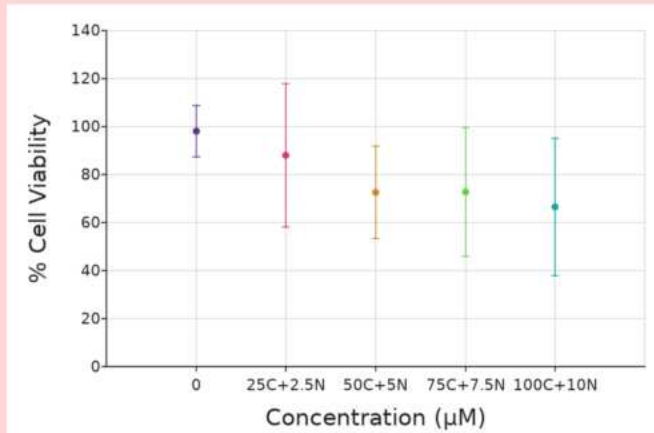


Figure 8. Graph representing cell viability in relation to the combined dosage concentration. Treatment groups were normalized to the control.

All Treatements

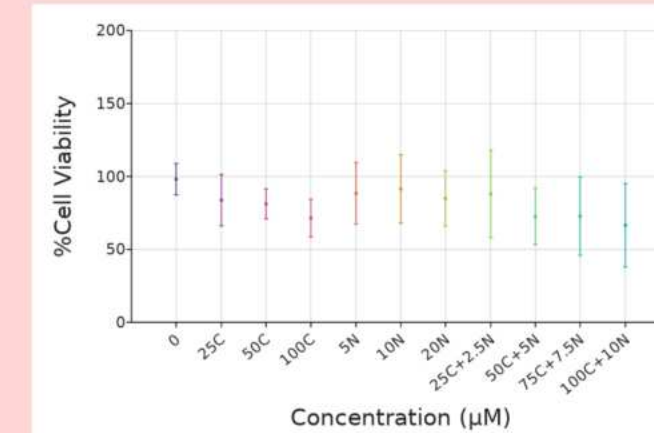


Figure 9. Graph representing cell viability in relation to the both the individual treatments and the combined dosage concentration. Treatment groups were normalized to the control.

Discussion

- Contrary to hypothesis:** Combined Curcumin and Nimbolide treatments did not decrease B16-F10 melanoma cell viability synergistically; no statistically significant differences were observed between combined treatments and their corresponding Curcumin-only groups
- Although several higher-dose combination treatments significantly reduced viability relative to untreated controls, the absence of significant differences between combination and single-agent Curcumin treatments suggests that Nimbolide did not enhance the anti-proliferative effects of Curcumin under the conditions tested
- May reflect insufficient Nimbolide potency/concentration, overlapping mechanisms of action, or high experimental variability masking synergistic effects

Synergy-Relevant Comparison	P-value	Interpretation
25C vs 25C+2.5N	1	No evidence combination improved effect
50C vs 50C+5N	1	No evidence combination improved effect
100C vs 100C+10N	1	No evidence combination improved effect

Table 1. Significant Kruskal-Wallis and post Dunn-Hoc test in determining a statistical difference in % Cell Viability between combined and individual treatments.

Future Work

- The current findings provide a foundation for further investigation. Multiple avenues remain for extending this work:
- Expanding the concentration range of Curcumin and Nimbolide to generate full dose-response curves and calculate combination index values for formal synergy analysis
 - Measuring p53 protein expression through Western blot analysis to determine whether reduced cell viability is associated with activation of tumor-suppressive pathways
 - Evaluating downstream effects on oncogenic signaling pathways, including MAPK, PI3K/AKT, apoptosis-related proteins, and oxidative stress pathways
 - Testing alternative treatment ratios or exposure times to determine whether synergy occurs under different dosing conditions

Key References

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